Logic Gates

Recall that digital circuits operate with binary logic levels. (high/low, 1/0, etc.)

These circuits, combinatorial logic systems, deal with boolean inputs and outputs.

We can map out these inputs and outputs with a truth table, representing all possible inputs and their corresponding outputs.

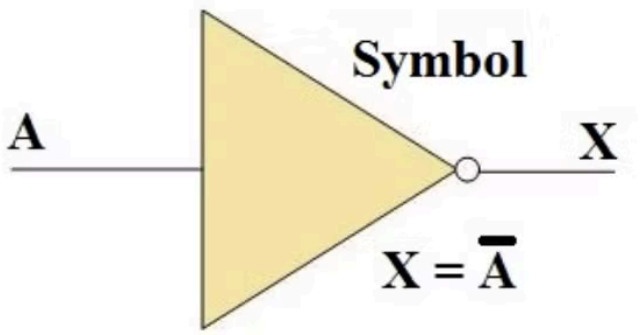
For example, an inverter/not gate has the truth table:

A	X
0	1
1	0

Base Logic Gates

- NOT gate
  - a NOT gate accepts one input and inverts the input.
  - The symbol is a bar on top, written as

$$\bar{A} = X$$

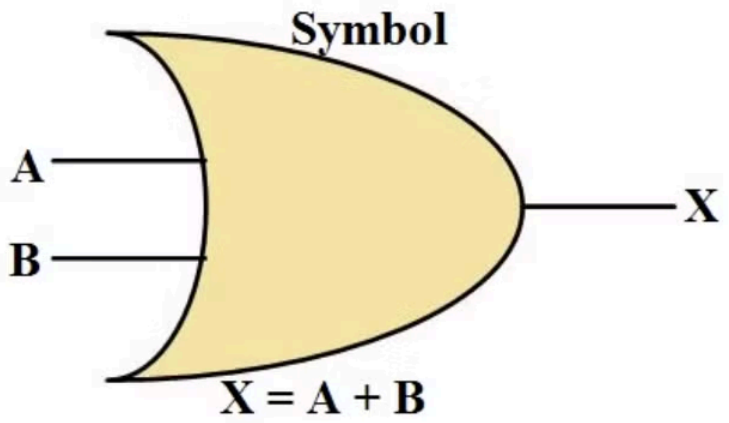


Truth Table

A	X
0	1
1	0

- OR gate
  - an OR gate accepts two inputs and gives one output when either input is 1.
  - The boolean algebra symbol for it is +, written as

$$A + B = X$$



Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Some properties of the OR gate:

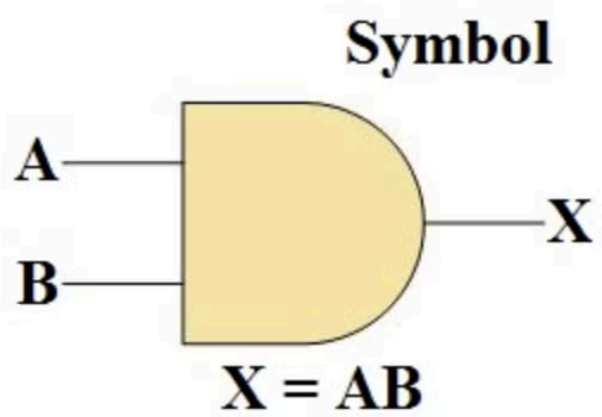
$$A + 0 = A$$
$$A + 1 = 1$$
$$A + A = A$$
$$A + \bar{A} = 1$$

AND gate

an AND gate accepts two inputs and gives one output when both inputs are 1.

The symbol is a multiplication dot, written as

$$A \cdot B = X \text{ or } AB = X$$



Truth Table

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

Some properties of the AND gate:

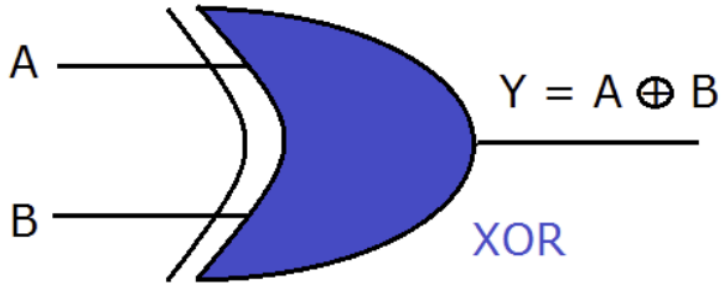
$$A \cdot 0 = 0$$
$$A \cdot 1 = A$$
$$A \cdot A = A$$
$$A \cdot \bar{A} = 0$$

XOR gate

a XOR gate has 2 inputs and gives one output when only one input is on.

The symbol and notation is as so:

$$A \oplus B = X$$



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Some properties of the XOR gate:

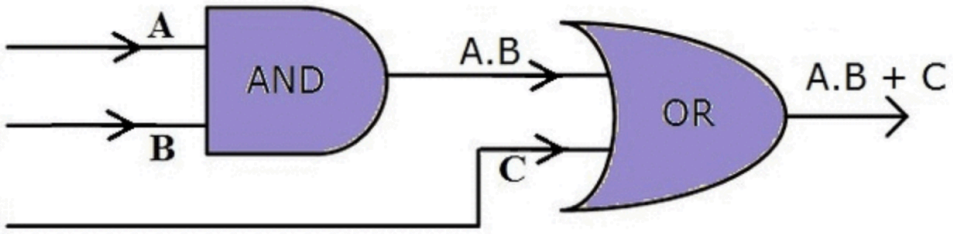
$$A \oplus 0 = A$$
$$A \oplus 1 = \bar{A}$$
$$A \oplus A = 0$$
$$A \oplus \bar{A} = 1$$

One thing to note is that the XOR gate can be represented as:

$$A \oplus B = \bar{A}B + A\bar{B}$$

Boolean Algebra

When multiple logic gates are combined to form a logic circuit, we can use boolean algebra notations to form a boolean expression that represents the logic circuit.

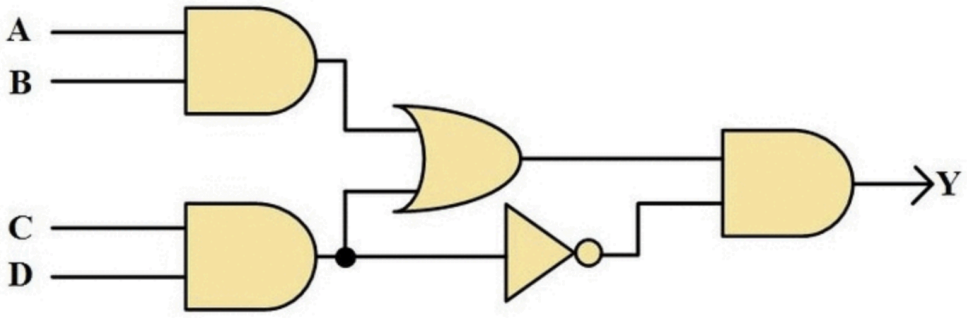


For example, for this circuit, we can represent the output as $AB + C$.

We can then use the boolean expression to determine the output given some inputs.

<u>Example</u>

For the circuit shown below, evaluate the output for A = 1, B = 1, C = 0 & D = 1.



After finding the boolean expression: $(AB + CD)\overline{CD}$, we can substitute in their values as so:

$$(AB + CD)\overline{CD} = (1 \cdot 1 + 0 \cdot 1)\overline{(0 \cdot 1)}$$

$$= (1 + 0) \cdot \bar{0}$$

$$= 1 \cdot 1$$

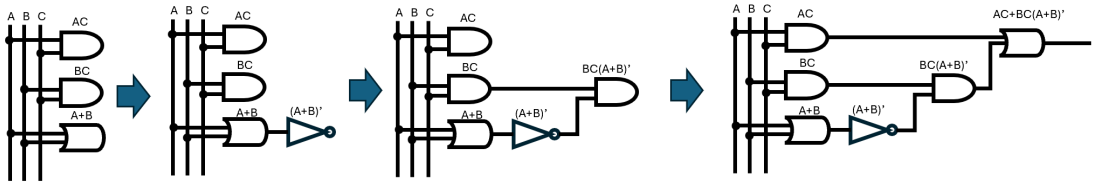
$$= 1$$

With a boolean expression, it can be implimented, i.e the logic circuit diagram drawn that represents the expression.

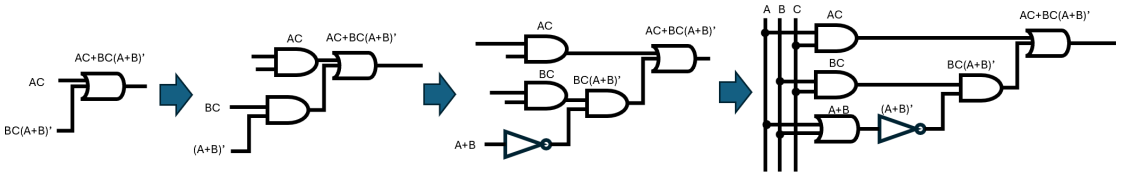
A boolean expression can be evaluated by either starting from the output and dealing with each term backwards, or by starting with the inputs

For example, implimenting $AC + BC\overline{A + B}$

We can start from the inputs by first making the most basic combination of the units (two inputs combined with one operation), then these values can be combined or inverted accordingly



With starting from the output, we take the main outer operation (OR) and split each of its inputs and keep splitting the inputs down to the original inputs



Note how junctions MUST be notated with a dot, and cross overs are not notated at all

Universal Gates (NOR, NAND)

the logic gates NOR and NAND are known as universal gates, as any digital circuit can be made with only these gates.

NOR

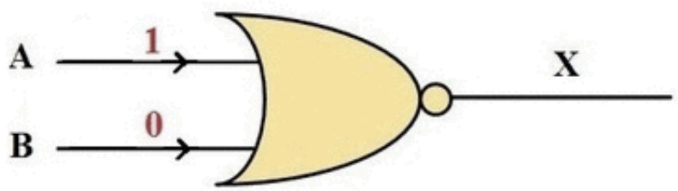
the NOR gate gives an output only when both inputs are 0. (inverse of OR, OR gives no input when both are 0)

NOR gate

Truth-Table

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

Symbol



Boolean Equation $X = \overline{A + B}$

The NOR gate can be thought of as an OR gate fed into a NOT gate (hence its boolean algebra notation)

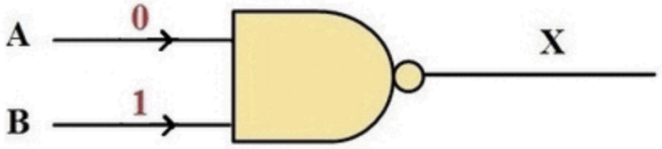
NAND

the NAND gate gives an output when either no or only one of the inputs are on. (inverse of AND, AND gives no input when no inputs or only one input)

Truth-Table

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

Symbol



$X = \overline{AB}$

Boolean Equation

The NAND gate can be thought of as an AND gate fed into a NOT gate.

Notice how both NAND and NOR have a circle in front, showing that a not gate is added/the OR/AND gate is inverted.

This means that we can also have a XNOR gate, where its a XOR fed into a NOT gate.